Acute effects of air pollution on paediatric asthma exacerbation: evidence of association and effect modification

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Introduction

Recent epidemiological research focuses on the health effects of air pollution (especially gaseous pollutants that have received less attention so far) on potentially sensitive population subgroups such as children. In the same time, there is growing demand for investigation of the possible factors modifying any observed effects. We investigated the acute effects of particulate matter with aerodynamic diameter < 10 μ (PM_{10}) and several gaseous pollutants, namely sulphur dioxide (SO_2), nitrogen dioxide (NO_2) and ozone (O_3), on emergency hospital admissions for asthma-related diagnosis among children aged 0-14 years in Athens, Greece. We explored possible modification patterns by season, gender and age. In addition, Athens, due to its metropolitan setting and geographical position, provides an opportune case for the investigation of the health effects of windblown desert dust particles, which originate mainly from the Sahara area. Hence, we also investigated whether the presence of windblown Sahara dust modifies the association between PM_{10} and paediatric hospital admissions for asthma.

Data and Methods

We collected data on daily counts of paediatric asthma emergency admissions in the three main Children's Hospitals, covering 85% of paediatric beds of Athens, from 2001 to 2004. There were 3,601 admissions for acute asthma-related diagnoses, with a daily mean of 3. 63% of admissions occurred among males and 72% among children aged less than 5 years. Asthma admissions revealed a distinct seasonal pattern with higher numbers during winter and lowest during summer when children leave the urban area for summer holidays. Daily air pollution measurements were provided by the monitoring network (Table 1). We identified 110 desert dust days between 2001-2004, using back-trajectory analysis to locate air mass transport in combination with a data driven criterion, based on high particles' concentrations.

Table 1. Distribution of air pollutants in Athens, Greece for 2001-2004.

	Mean	Median	IQR[a]	Total No of days > WHO guideline
PM_{10}-24h (μg/m^3)	43.9	41.0	31.1-52.7	449 (>50 μg/m^3)
SO_2-24h (μg/m^3)	16.8	14.1	9.0-22.0	438 (>20μg/m^3)
NO_2-1h max (μg/m^3)	84.8	81.4	64.2-101.5	2 (>200 μg/m^3)
O_3-8h (μg/m^3)	70.9	69.5	48.6-91.9	251 (>100 μg/m^3)

[a] Inter-quartile range.

The pollution-admissions associations were investigated using Poisson regression models with natural splines to control for possible confounding by season, meteorology, day of the week, influenza epidemics and holiday effect. The levels of the pollutant under investigation on the same day (lag 0) with the admission were included in the model. To examine whether the effects of air pollution on paediatric asthma admissions are spread over several days, we fitted distributed lag models including the levels of the pollutants the same day and up to 2 days before the admission. We investigated possible effect modification by season, gender and age group using stratification by these potential modifiers. Finally, we investigated the effect of dust events on mortality using an indicator variable and its interaction with PM_{10} concentrations.

Results

Table 2. Percent Increase and 95% Confidence Intervals (95%CIs) in daily asthma admissions for ages 0-14 years in Athens, Greece for 2001-2004, for 10 μg/m3 increase in the levels of the corresponding pollutant in the same day (lag 0).

	Annual	Winter	Spring	Summer	Autumn
PM_{10}	2.54 (0.06,5.08)	5.64 (0.16, 11.41)	2.85 (-0.46, 6.26)	6.71 (-4.55, 19.30)	3.87 (-1.48, 9.51)
SO_2-24h	5.98 (0.88, 11.33)	7.25 (0.27, 14.7)	11.06 (1.85, 21.10)	15.64 (-5.48, 41.49)	14.44 (12.89, 16.02)
NO_2-1h	1.10 (-0.68, 2.91)	2.18 (-1.21, 5.67)	2.28 (-0.79, 5.45)	4.57 (-0.61, 10.02)	2.27 (-1.43, 611)
O_3-8h	-3.07 (-6.55, 0.53)	-4.94 (-11.07, 1.60)	-4.76 (-10.34, 1.16)	9.30 (-1.07, 20.76)	-7.72 (-13.96, -1.03)

All pollutants had higher effects on paediatric asthma during the summer period, which were statistically non-significant due to the small number of events (Table 2). SO_2 was the pollutant with the highest effect on asthma admissions, with strong statistically significant effects during autumn. PM_{10} was associated with a statistically significant increase in admissions overall, attributed mainly to the effects of particles during winter.

The association with NO_2 was positive though non significant in all analyses. O_3 only had an effect on children's asthma admissions during summer when the association was positive and indicative.

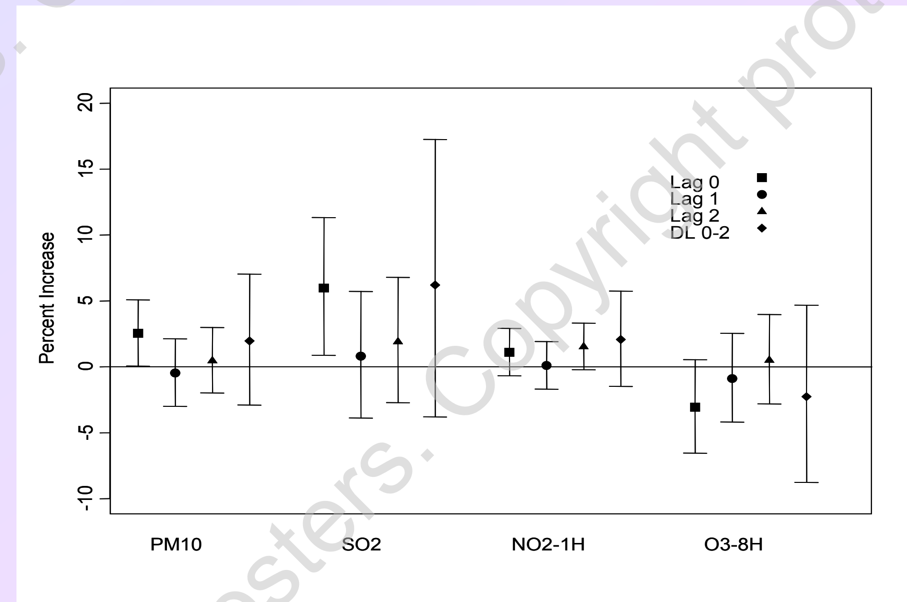

Figure 1. Percent Increase and 95% Confidence Intervals in daily asthma admissions for ages 0-14 in Athens, Greece in 2001-2004 associated with 10 μg/m^3 increase in the levels of the pollutant for different exposure lags.

When we investigated the effect modification patterns by gender and age group (0-4 years and 5-14 years old), we found a clear pattern for gender while there was a less pronounced effect of age. More specifically, air pollution effects were encountered in males, while the effects on females were practically null. The effects of PM_{10}, SO_2 and summer O_3 on paediatric asthma were higher among those aged 5-14years but non-statistically significant due to the smaller counts, except for O_3 that was associated with a statistically significant increase of 21.25% (95%CIs: 2.32%, 43.68%) for 10 μg/m^3 increase during summer.

Finally, we found higher effect of particles on asthma admissions during days with desert dust, since during non-desert days a 10μg/m^3 increase in PM_{10} resulted in 2.06% (95%CI: -1.01%, 5.21%) increase in admissions while during desert dust days the effect increased to 4.12% (95%CI: 0.11%, 8.30%).

For PM_{10} and SO_2 the highest effect was observed in the same day as the admission while there appeared to be a lagged effect in the response to NO_2 and O_3 (Figure 1).

Conclusions

- We found strong adverse effects of exposure to particulate pollution and SO_2 on emergency hospital admissions for asthma-related diagnosis among children aged 0-14 years. The associations persisted after mutual adjustment, indicating independent effects.
- Our analyses point to significantly elevated effects of particles during desert dust days and are suggestive of ozone effects at higher concentrations encountered during the summer.

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